

XGBoost

XGBoost Introduction

XGBoost (extream Gradient Boosting) is a highly efficient, scalable & popular open-source machine Learning library designed for supervised learning, particularly building high performance model using decision tree ensembles.

It optimizes gradient boosting frameworks by utilizing parallel processing and regularization techniques to prevent overfitting, making it a go to algorithm for structured / tabular data.

How XGBoost Model Works

1) Ensemble Learning

It combines predictions from many "weak" learners (usually classification and regression trees or CART) to create one strong predictive model.

2) Boosting Mechanism

Trees are built sequentially, where each new tree corrects the residual errors (residual learning) made by the previous ensemble of trees.

3) Gradient Descent :

XGBoost uses a gradient descent algorithm to minimize errors (loss function) when adding new models, allowing it to rapidly improve accuracy.

4) Regularization

Unlike traditional GBM, XGBoost includes L_1 (Lasso) and L_2 (ridge) regularization to control model complexity and reduce overfitting.

* XGBoost Features

1) Flexibility

1) Cross-Platform

can run into any platform (Linux, Windows, mac)

2) Multiple Language Support

Supports multiple programming language

2) Integration with other libraries and tools

4) Support all kinds of ML problems.

2) Speed

- 1) Parallel Processing
- 2) Optimized Data Structure
- 3) Cache awareness
- 4) Out of core computing
- 5) Distributed computing
- 6) GPU Support